

# OMAR BAHAAELDIN

Data Scientist · Machine Learning · Forecasting & Predictive Analytics

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## PROFESSIONAL SUMMARY

AI & Data Science graduate (ESLSCA University) with a project portfolio spanning predictive modeling, time-series forecasting, classification, and information retrieval. Built a stock market forecasting model where a Transformer architecture reduced MAE/RMSE by ~20% over ARIMA/SARIMA baselines, and a credit default risk classifier achieving 0.91 AUC on imbalanced financial data. Proficient across the full data science workflow: EDA, feature engineering, model selection, evaluation, and actionable interpretation. Targeting data scientist or analyst roles in finance, business intelligence, or product analytics.

## TECHNICAL SKILLS

|                              |                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Modeling</b>              | Scikit-learn · LightGBM · ARIMA · SARIMA · LSTM · Transformers · BERT · Sentence Transformers · PyTorch              |
| <b>Data Engineering</b>      | Pandas · NumPy · ETL pipelines · Feature engineering · SMOTE · EDA · Time series analysis                            |
| <b>Evaluation</b>            | MAE · RMSE · R <sup>2</sup> · AUC · F1 · MAP · nDCG@10 · Ablation studies · Cross-validation · Hyperparameter tuning |
| <b>Databases</b>             | SQL · PostgreSQL · MySQL                                                                                             |
| <b>Languages &amp; Tools</b> | Python · JavaScript · Docker · Linux · Git                                                                           |

## PROJECTS

- Stock Market Forecasting — S&P 500** | ARIMA · SARIMA · LSTM · Transformers · Pandas Jun 2025
- Comparative study of statistical (ARIMA/SARIMA) and deep learning (LSTM, Transformer) forecasting methods on S&P 500 time-series data.
  - Transformer model reduced forecasting error by ~20% over ARIMA/SARIMA baselines across MAE, RMSE, and R<sup>2</sup> metrics.
  - Full pipeline: stationarity testing, data transformation, hyperparameter tuning, walk-forward evaluation, and model comparison.
- Credit Default Risk Prediction** | LightGBM · Scikit-learn · SMOTE · Feature Engineering 2024
- Binary classification on imbalanced financial dataset — achieved 0.91 AUC via SMOTE oversampling, feature engineering, and systematic hyperparameter tuning.
  - Demonstrates applied ML methodology: class balancing, feature importance analysis, and model validation on real-world financial data.
- Information Retrieval Evaluation System** | BERT · Sentence Transformers · BM25 · MAP · nDCG@10 2024
- Benchmarked semantic (BERT) vs. keyword retrieval (BM25) — BERT outperformed by 18% MAP and 14% nDCG@10.
  - Rigorous evaluation framework using standard IR metrics — relevant for search ranking, recommendation, and NLP product analytics.
- Fabric Defect Detection (Applied ML)** | PyTorch · YOLOv11/v12 · RT-DETR · OpenCV Jun 2025
- Systematic model evaluation across four architectures; achieved 61.3% mAP@50 and 62.7% F1 — demonstrates quantitative methodology applicable to any structured ML problem.
  - Ablation study across architectures mirrors rigorous model selection practices used in data science workflows.
- Waza3ly — Data-Backed Platform** | PHP · PostgreSQL · REST APIs 2024
- Full-stack platform with relational schema designed to support donor, volunteer, and delivery analytics — demonstrates ability to structure data for downstream analysis.

## EXPERIENCE

- Full-Stack Web Developer** | SISCOM Jan 2023 UAE · Remote
- Sole developer for 70+ page production website — drove 73.3% uplift in online visibility and 1,000+ monthly visitors.
  - Built and shipped a full technical system independently — same discipline applied to data science project work.

## EDUCATION

**B.Sc. AI & Data Science** | ESLSCA University (Paris Business School)  
Jul 2025 · Egypt  
American High School Diploma — GPA 3.8 | Pioneers International School, Giza

CERTIFICATIONS & RECOGNITION

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|-------------------------------------------------------------------------------|----------|
| Cisco Networking Academy — Network Support & Security                         | Oct 2025 |
| EVAi AI Hackathon — Outstanding Performance Award   Coach Academy / EVA Group | 2024     |